

Andong Deng

Ph.D. Candidate @ University of Central Florida | Multimodal AI

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RESEARCH INTERESTS

My research focuses on multimodal AI, particularly on building models that can perceive and reason over complex visual dynamics across different modalities. I develop benchmarks to uncover the limitations of current MLLMs in compositional and long-horizon reasoning, and design frameworks that enable structured spatiotemporal understanding. My long-term goal is to develop multimodal intelligence to interact with the physical world.

Research Areas: Computer Vision, Multimodal AI, AI4Science

EDUCATION

University of Central Florida, U.S. *Sep. 2022 - Present*

Ph.D. in Computer Science

Advisor: Dr. Chen Chen

Shanghai Jiao Tong University, China *Sep. 2018 - Mar. 2021*

Master of Engineering

Advisor: Dr. Weiwei Cai

Sichuan University, China *Sep. 2013 - Jun. 2017*

Bachelor of Energy and Power Engineering

EXPERIENCE

Research Scientist Intern, ByteDance, San Diego *May 2026 - Present*

Worked with: Dr. Kai Zhang (kai.zhang@bytedance.com)

- Efficient Multimodal LLMs

Research Scientist Intern, ByteDance, San Jose *May 2025 - May 2026*

Worked with: Dr. Sijie Zhu (sijiezhu@bytedance.com) and Dr. Longyin Wen (longyin.wen@bytedance.com)

- Real-world video editing benchmark (work published in CVPR Findings 2026)

- Multi-video reasoning and VLM post-training (in submission)

Research Scientist Intern, United Imaging Intelligence (UII America), Boston *May 2024 - Nov. 2024*

Worked with: Dr. Zhongpai Gao (gaozhongpai@gmail.com), Dr. Benjamin Planche (benjamin@planche.me) and Dr. Ziyang Wu (ziyan.wu@uii-ai.com)

- Video LLM for temporal grounding (work published in CVPR 2025)

Research Intern, Baidu Research, Big Data Lab, Beijing *Jul. 2021 - Mar. 2022*

Worked with: Dr. Xingjian Li (lixingjian@baidu.com) and Dr. Dejing Dou (doudejing@baidu.com)

- Self-supervised and transfer learning (work published in ICCV 2023)

Research Assistant, Renmin University of China, Beijing *Apr. 2021 - Jun. 2022*

Worked with: Dr. Di Hu (dihu@ruc.edu.cn)

- Audio-visual video understanding (work published in CVPR 2022)

Research Intern, SenseTime, Shanghai *May 2020 - Jun. 2021*

Worked with: Dr. Yang Cao (caoyang@sensetime.com) and Dr. Rui Zhao (zhaorui@sensetime.com)

- Representation learning and video generation

PUBLICATIONS

1. **Andong Deng**, Taojiannan Yang, Shoubin Yu, Lincoln Spencer, Mohit Bansal, Chen Chen, Serena Yeung-Levy, and Xiaohan Wang. SciVideoBench: Benchmarking Scientific Video Reasoning in Large Multimodal Models. ICML 2026, *Best Paper Award, ICCV KnowledgeMR Workshop 2025*
2. **Andong Deng**, Dawei Du, Zhenfang Chen, Wen Zhong, Fan Chen, Guang Chen, Chia-Wen Kuo, Longyin Wen, Chen Chen, and Sijie Zhu. VEBench: Benchmarking Large Multimodal Models for Real-world Video Editing. CVPR Findings, 2026
3. **Andong Deng**, Zhongpai Gao, Anwesa Choudhuri, Benjamin Planche, Meng Zheng, Bin Wang, Terrence Chen, Chen Chen, Ziyang Wu. Seq2Time: Sequential Knowledge Transfer for Video LLM Temporal Grounding, CVPR 2025
4. **Andong Deng**, Tongjia Chen, Shoubin Yu, Taojiannan Yang, Lincoln Spencer, Yapeng Tian, Ajmal Saeed Mian, Mohit Bansal, and Chen Chen. Motion-Grounded Video Reasoning: Understanding and Perceiving Motion at Pixel Level, CVPR 2025
5. **Andong Deng***, Taojiannan Yang* and Chen Chen. A Large-scale Study of Spatiotemporal Representation Learning with a New Benchmark on Action Recognition, ICCV 2023
6. **Andong Deng***, Xingjian Li*, Di Hu, Tianyang Wang, Haoyi Xiong, and Cheng-Zhong Xu. Towards Inadequately Pre-trained Models in Transfer Learning, ICCV 2023
7. **Andong Deng**, Taojiannan Yang, Chen Chen, Qian Chen, Leslie Neely, Sakiko Oyama. Language-assisted Deep Learning for Autistic Behaviors Recognition, Smart Health, 2023
8. Shayda Moezzi, Umer Saleem, **Andong Deng**, Chen Chen, and Sarah Ostadabbas. Position: Video LLMs Must Not Ignore the Pixel Dynamics in Plain Sight. ICML Position Track, 2026.
9. Haopeng Li, **Andong Deng**, Jun Liu, Hossein Rahmani, Yulan Guo, Bernt Schiele, Mohammed Bennamoun, and Qihong Ke. Sports-QA: A Large-Scale Video Question Answering Benchmark for Complex and Professional Sports, International Journal of Computer Vision, 2026
10. Yuanhao Zou, Shengji Jin, **Andong Deng**, Youpeng Zhao, Jun Wang, and Chen Chen. AIR: Enabling Adaptive, Iterative, and Reasoning-based Frame Selection For Video Question Answering, ICLR 2026
11. Bin Wang, Anwesa Choudhuri, Meng Zheng, Zhongpai Gao, Benjamin Planche, **Andong Deng**, Qin Liu, Terrence Chen, Ulas Bagci, Ziyang Wu. Order-aware Interactive Segmentation, ICLR 2025
12. Shoubin Yu, Zhongying Zhao, Haoshu Fang, **Andong Deng** and Cewu Lu. Regularity Learning via Explicit Distribution Modeling for Skeletal Video Anomaly Detection, IEEE Transactions on Circuits and Systems for Video Technology, 2023
13. Wenke Xia, Xingjian Li, **Andong Deng**, Haoyi Xiong, Dejing Dou, and Di Hu. Robust Cross-Modal Knowledge Distillation for Unconstrained Videos, ICME 2023
14. Xiaokang Peng*, Yake Wei*, **Andong Deng**, Dong Wang, and Di Hu. Balanced Multimodal Learning via On-the-fly Gradient Modulation, CVPR 2022, *Oral Presentation*

PRE-PRINTS

1. **Andong Deng**, Zhenfang Chen, Dawei Du, Wen Zhong, Fan Chen, Guang Chen, Longyin Wen, Chen Chen, Sijie Zhu. VideoInspector: Agentic Multi-Video Reasoning with Compositional Segment Training.

AWARDS

Best Paper Award, ICCV KnowledgeMR Workshop, 2025

PROFESSIONAL SERVICES

Conference Reviewing

- **Computer Vision:** CVPR (2022, 2024, 2025, 2026), ICCV (2023, 2025), ECCV (2022, 2026)
- **Machine Learning:** NeurIPS (2025, 2026), ICLR (2025, 2026), ICML (2026)

Journal Reviewing

- International Journal of Computer Vision (IJCV)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)